Fiscal Year:	FY 2023	**Task Last Updated:**	FY 02/06/2023
PI Name:	Quintiere, James Ph.D.		
Project Title:	Experimental Investigation of Emulated Burning Rate at Various Gravity Levels		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20742-3031	**Congressional District:**	5
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	01/01/2015	**End Date:**	02/08/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
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Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 02/08/2024 per NSSC information (Ed., 2/24/23) NOTE: Extended to 02/08/2023 per NSSC information (Ed., 4/11/22) NOTE: Extended to 02/08/2022 per NSSC information (Ed., 12/3/21) NOTE: New grant awarded 80NSSC20M0119 with new start/end dates 4/9/2020-10/08/2021; added on to existing grant NNX15AD06A per D. Stocker/GRC (Ed., 10/13/20) NOTE: Extended to 12/31/2020 per NSSC information (Ed., 11/7/19) NOTE: Extended to 12/31/2019 per NSSC information (Ed., 2/3/17)		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):	Sunderland, Peter Ph.D. (University of Maryland) deRis, John Ph.D. (FM Global)		
Grant/Contract No.:	80NSSC20M0119 ; NNX15AD06A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: New grant awarded 80NSSC20M0119 with new start/end dates 4/9/2020-10/08/2021; added on to existing grant NNX15AD06A per D. Stocker/GRC (Ed., 10/13/20) NOTE: Continuation of "Experimental Investigation of Emulated Burning Rate at Various Gravity Levels," grant NNX10AD98G, with the same Principal Investigator (PI) (Ed., 1/14/2015) A novel experimental apparatus, the Burning Rate Emulator (BRE), was developed. This apparatus is capable of assessing ignitability, steady burning rate, and extinction characteristics of virtual solid and liquid fuels as a function of gravity level, flow characteristics, and orientation. By changing the fuel gas and its flow rate, it enables the emulation of steady combustion for a range of liquid and solid fuels having B numbers (mass transfer numbers) and radiation characteristics that fall anywhere within physically realistic bounds. Measurements of the critical mass flux at ignition can also be made. These measurements were conducted as a function of scale and orientation to assess turbulent and laminar burning in natural convection. The experiment is well suited to testing in the Combustion Integrated Rack aboard the International Space Station. The Burning Rate Emulator (BRE) simulated the steady burning of liquids and solids using a rectangular porous burner fed by gaseous fuel and equipped with a total heat flux sensor. BRE controlled the fuel flow rate based on the measured heat flux at the burner and the measured surface temperature. This allowed it to use a small number of gaseous fuels to match the following four properties of a large number of condensed fuels: heat of combustion (LHV), heat of vaporization (L), surface temperature (Ts), and laminar smoke point. Extinguishment was considered via changes in fuel flow rate. Both low flow rates (for kinetic extinction) and high flow rates (for radiative extinction) were considered. The BRE flight program seeks to establish the burning conditions in a quiescent microgravity ambient in terms of material properties. The specific flight objectives are as follows. 1. Observe burning behavior in long-duration microgravity for CH4/N2 and C2H4/N2 mixtures on porous circular burners of 25 and 50 mm diameter. Consider O2/N2 ambients with various pressures and ambient O2 concentrations and pressures. NASA-considered atmosphere for human space missions will be examined: 14.7 psia, 21% oxygen: 10.2 psia, 26.5%; and 8.2 psia, 34% oxygen. Emphasize fuel conditions that can be related to a diverse range of condensed fuels. 2. Consider the conditions of Objective 1 for ambients of O2/CO2 (desired). 3. Consider the conditions of Objectives 1 – 2 for extinction limits.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project is expected to improve the understanding of fire and combustion processes in both microgravity and normal gravity. It also could advance the state of the art in the design of porous burners and simple heat flux meters.
Task Progress:	We continued working on the data analysis and publication of all the Burning Rate Emulator (BRE) data. One dissertation, one journal paper, and one conference paper were published this year. [Ed. Note: See Bibliography.] Dehghani P, De Ris JL, Quintiere JG. "Demonstrating steady burning for small flat materials in microgravity in a quiescent ambient." Proceedings of the Combustion Institute. 2022 Nov 8. Online ahead of print. https:// , Nov-2022 The likelihood of steady burning in microgravity is examined with BRE using a gas burner with a flat 25 mm diameter porous surface with two embedded heat flux sensors. The data are greatly expanded from a previous work. The fuel mixture used in this analysis are ethylene and ethylene diluted with nitrogen, resulting in a range of heats of combustion from 12 to 47.2 kJ/g. An analytical solution of these flames was utilized to compute their steadiness, temperatures at flame sheet, and sizes. 103 flames were ignited in an ambient with nominal oxygen mole fractions of 0.21, 0.265, 0.34, and 0.40. 49 of those flames burned for the entire duration of the test before the fuel supply was terminated, while the remainder self-extinguished. The flames attained at least 90% of their asymptotic heights for all 49 fuel terminated flames. An estimated critical steady flame temperature ranging from 1100 to 1200 K was observed, segregating the sustained and self-extinguished flames. Flame temperatures for the 49 terminated flames were above the critical temperature range, indicating sustained steady burning, whereas below they self-extinguished. An empirical correlation of the radiative fraction, based on radiation theory with O, a dimensionless parameter, was used in prediction for real fuels at steady burning in microgravity. A flammability diagram, as a plot of the emulated fuel heat of gasification with its burning mass flux, for the range of steady data, was presented in technical reporting to NASA, comparing how real fuels might burn steadily in normal and microgravity conditions. The results suggest that fuel mass burning rates would be about 3 to 4-times higher in 1-g. The heats of gasification and combustion of some real fuels were presented and compared against the burner flames.
Bibliography Type:	Description: (Last Updated: 10/08/2025)
Articles in Peer-reviewed Journals	Dehghani P, De Ris JL, Quintiere JG. "Demonstrating steady burning for small flat materials in microgravity in a quiescent ambient." Proceedings of the Combustion Institute. 2022 Nov 8;39(3):3949-58. https://doi.org/10.1016/j.proci.2022.08.107 , Nov-2022
Papers from Meeting Proceedings	Dehghani P, Quintiere JG. "Using a gas supplied burner to emulate condensed phase fuels in microgravity." 9th Fuel and Combustion Conference of Iran, Virtual, February 8-10, 2022. Abstracts. 9th Fuel and Combustion Conference of Iran, Virtual, February 8-10, 2022. , Feb-2022